

SQL Data Analysis — Project 3

Industrial Training Kit | DecodeLabs | Batch 2026

Overview

This project performs SQL Data Analysis on the cleaned e-commerce orders dataset (1,200 records) produced in Project 1. The dataset is loaded into a SQLite database and queried using 10 structured SQL queries — covering SELECT, WHERE, ORDER BY, GROUP BY, HAVING, COUNT, SUM, and AVG — to extract real business insights through relational logic.

Key Concept

SQL is a **declarative language** — you define *what* you want, and the database engine decides *how* to fetch it. Understanding the logical execution order (FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY) is critical to writing correct queries and avoiding the common "Alias Trap".

Queries Written

#	Query	Clauses Used	Business Question Answered
1	Preview First 10 Orders	SELECT, LIMIT	What does the data look like?
2	All Delivered Orders	WHERE, ORDER BY	Which delivered orders had highest value?
3	High-Value Orders > \$2000	WHERE, ORDER BY	Which orders are flagged as high-value?
4	Orders by Product	COUNT, GROUP BY	Which product is ordered most?
5	Revenue by Product	SUM, GROUP BY	Which product generates the most revenue?
6	Avg Order Value by Payment Method	AVG, GROUP BY	Which payment method drives highest order value?

#	Query	Clauses Used	Business Question Answered
7	Orders & Revenue by Referral Source	COUNT, SUM, AVG, GROUP BY	Which channel brings the most customers and revenue?
8	High-Volume Products only	GROUP BY, HAVING	Which products exceed 170 orders?
9	Cancelled Orders by Product	WHERE, GROUP BY, COUNT	Which product has the most cancellations?
10	Revenue Contribution % per Product	SUM, Subquery, GROUP BY	What percentage of total revenue does each product generate?

Key Findings

- **Chair and Printer** are nearly tied for highest revenue (~\$195,600 each), despite Printer leading in order count.
- **Credit Card** has the highest average order value (\$1,127.55) among all payment methods.
- **Instagram** drives the most orders (259) and the most total revenue (\$275,285) — the top acquisition channel.
- **Chair** has the most cancellations (45 orders) — worth investigating for a product or fulfillment issue.
- Revenue is fairly evenly distributed: each product contributes between 12% and 15.5% of total revenue.

Tech Stack

- Python 3
- SQLite3 (via Python's built-in `sqlite3` module)
- Pandas (for loading CSV → database and displaying results)

Repository Structure

```
|— sql_analysis.py      # Main SQL query
script
|— orders.db           # SQLite
database (auto-generated from cleaned CSV)
```

```
|— sql_query_log.csv      # Log of all
queries run
|— cleaned_dataset.csv   # Input dataset
(output of Project 1)
|— README.md
```

How to Run

```
pip install pandas
python sql_analysis.py
```

This will create `orders.db` from the cleaned CSV, run all 10 queries, print results to the console, and save `sql_query_log.csv`.

SQL Execution Order (Key Concept from the Kit)

The database does NOT read your query top to bottom. The actual order is:

1. FROM → Locate the table
2. WHERE → Filter individual rows
3. GROUP BY → Bucket the filtered rows
4. HAVING → Filter the buckets
5. SELECT → Pick columns and calculate
6. ORDER BY → Sort the final output

This is why you cannot use a SELECT alias inside a WHERE clause — the alias doesn't exist yet when WHERE runs.

*Part of the DecodeLabs Industrial Training Kit — Data Analytics Track, Batch 2026.
Builds on Project 1 (Data Cleaning) and Project 2 (EDA).*